Difference-in-Differences Estimates under Selective Migration*

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Abstract

Difference-in-differences designs often study place-based treatments that can trigger migration in and out of treated areas. When treatment changes who is observed, aggregate and within-individual DiD no longer retrieve the average treatment effect on the treated. This paper uses principal stratification to characterize three estimands under treatment-induced migration: a locality-level treatment effect, a stayer average treatment effect, and an individual ATT for the pre-treatment population. Aggregate DiD identifies the locality-level effect under aggregate parallel trends, while within-individual DiD identifies the stayer effect under stayer parallel trends. Together, the two assumptions imply parallel trends for escapees, a non-trivial restriction on the group whose departure drives compositional change. With panel data, the stayer effect and the compositional term are point-identified, and the ATT lies in a one-parameter sensitivity region. With repeated cross-sections, Lee-type bounds apply in the one-sided exit case. An appendix extends these results to natural turnover.

Keywords: Difference-in-differences, migration, panel data, policy evaluation, principal stratification.
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Introduction

Difference-in-differences (DiD) is among the most widely used research designs in the social sciences, and a large methodological literature has refined the design to handle staggered adoption, heterogeneous treatment effects, and violations of parallel trends (for recent developments, see Roth et al. 2023; Abadie et al. 2025). A fundamental challenge arises, however, when the treatment itself changes the composition of the observed population. Many place-based treatments, such as policy changes, environmental shocks, and infrastructure investments, trigger migration across locations. Individuals selectively enter or exit the treated area, and the population observed after treatment differs systematically from the one observed before.

Applied researchers typically pursue one of two DiD designs. First, they estimate aggregate DiD at the locality level. This captures what happens to the locality on average, but it requires on stable unit treatment value (SUTVA) assumptions to infer behavioral responses. Second, they estimate within-individual DiD with individual fixed effects on the individuals observed in both periods. This keeps composition fixed by design by restricting attention to stayers, and therefore targets a selected subpopulation rather than the full pre-treatment population. Which causal quantities these two approaches identify under selective migration has thus far received limited scrutiny.

This paper treats the problem of selective migration as one of estimand rather than of bias (although see Sant'Anna and Xu 2025; Rathnayakea et al. 2024). It uses principal stratification (Frangakis and Rubin 2002) to formalize the quantities estimated by DiD in a two-period setting with treatment-induced migration. The paper first formally distinguishes among the locality-level treatment effect τ^{LOC}, the average treatment effect on the treated τ^{ATT} for the pre-treatment population, and the survivor average treatment effect τ^{SATE} for stayers. The key take-away here is that the locality-level effect need not be an average of individual treatment effects over a fixed population. Under treatment-induced migration, the place-based estimand and the person-based estimands diverge.

Next, the paper shows what the familiar DiD estimators recover in a setting without natural turnover, where composition changes come only from treatment-induced migration in treated areas and the control population is stable. In such setting, aggregate DiD identify the locality-level effect under aggregate parallel trends, but not the ATT. Meanwhile, within-individual DiD identify the stayer effect under stayer parallel trends. Taken together, the two parallel-trends assumptions also imply that would-be escapees share the control trend, a non-trivial joint restriction given that escapees are a selected subpopulation. Combined,

the two assumptions allow point identification of the compositional effect and a sensitivity analysis for the ATT, but they also implicitly rule out differential pre-treatment trends for the very group whose departure drives compositional change.

Finally, the paper describes identification options by data regime. With individual panel data, the SATE and the compositional effect are point-identified without natural turnover, while the ATT lies in a sensitivity region indexed by a single selection parameter. With repeated cross-sections, a clean Lee-type argument is available in the one-sided case where treatment induces exits but not entries. With aggregate data alone, only the locality-level effect is identified. The appendix relaxes the no-turnover restrictions and shows what changes once natural turnover is restored in both treated and control areas. Parallel trends assumptions become more composite, and several individual-level estimands lose point identification. However, two conditions, namely the absence of untreated exits from treated areas and a common natural turnover assumption across treated and control localities, are sufficient to recover the full no-turnover identification results.

This paper builds on recent applications of principal stratification to migration and place-based treatment settings by Marbach (2024, 2025). It also relates to recent work on DiD with compositional change. Sant'Anna and Xu (2025) derive efficient estimators under selection on observables for repeated cross-sections. Rathnayakea et al. (2024) study bounds under endogenous selection, and Ghanem et al. (2024) extend changes-in-changes arguments under rank restrictions. Lee (2009) provides the basic trimming logic for one-sided selection. Relative to this literature, the contribution here is to make the estimand distinction primary and to show how common DiD estimators line up with different causal questions once migration is part of the treatment response.

A related literature studies DiD under interference, where treatment in one area changes outcomes in another (Mealli and Viviens 2025; Xu 2025; Butts 2024). Those papers focus on behavioral spillovers. The present paper focuses on a different channel, namely that treatment changes who is observed in each locality even when outcomes in control areas are otherwise untouched. The two channels may operate together in practice, but it is useful to separate them analytically before combining them.

Framework

Consider agents $i \in \{1, \ldots, N\}$ observed over two periods $t \in \{0, 1\}$. Agents are located in localities partitioned into treated ($\mathcal{T}$) and control ($\mathcal{C}$) areas. Treatment $D_j \in \{0, 1\}$ is a policy or any absorbing-state shock assigned at the locality level and activated at $t = 1$.

In a standard DiD design without migration, the same agents are present in each locality in both periods. With a fixed population, the locality-level treatment effect, defined as the change in the locality mean caused by treatment, is simply the average of individual treatment effects over that population. The locality-level effect and the individual ATT are the same object. Similarly, aggregate DiD and within-individual DiD are interchangeable: both identify the ATT under standard parallel trends. Appendix A1 states these equivalences formally. However, when treatment triggers migration, these equivalences break down. The population observed after treatment differs from the one observed before, so a locality-level comparison is no longer an individual-level comparison. The rest of this section introduces the notation needed to formalize that distinction.

A key feature of place-based designs with migration is that an agent's outcome depends not only on the treatment status of the locality but also on whether the agent remains there. Let $Y_{it}(d, s)$ denote the potential outcome for agent i at time t when the locality treatment status is $d \in \{0, 1\}$ and the agent is present in the locality ($s = 1$) or absent from it ($s = 0$).

Because migration is itself a response to treatment, let $S_i(d) \in \{0, 1\}$ denote the potential staying decision of agent i under treatment status d, where $S_i(d) = 1$ means that the agent is present in the locality at $t = 1$ and $S_i(d) = 0$ otherwise. Let S_{it} denote the observed presence status of agent i at time t. For agents initially present in treated localities, $S_{i0} = 1$ by construction, and at $t = 1$ the realized presence status is $S_{i1} = S_i(D_j)$. The observed outcome satisfies consistency: for each agent present in the locality at time t, $Y_{it} = Y_{it}(D_j, S_{it})$.

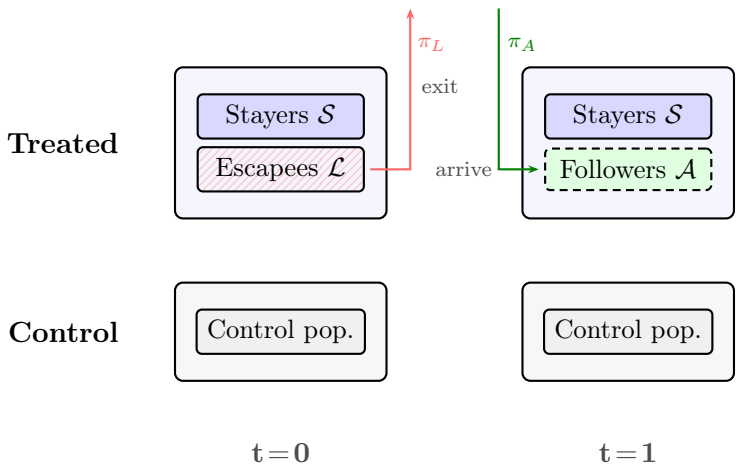
Principal strata

Following Frangakis and Rubin (2002), each agent connected to a treated locality belongs to one of four principal strata defined by the joint potential staying decisions $(S_i(0), S_i(1))$.

- **Stayers** $\mathcal{S}$: $(S_i(0), S_i(1)) = (1, 1)$, present regardless of treatment.
- **Escapees** $\mathcal{L}$: $(S_i(0), S_i(1)) = (1, 0)$, present under control and leaving under treatment.
- **Followers** $\mathcal{A}$: $(S_i(0), S_i(1)) = (0, 1)$, absent under control and arriving under treatment.
- **Never-present** $\mathcal{N}$: $(S_i(0), S_i(1)) = (0, 0)$, absent regardless of treatment.

In the no-turnover setting, all agents initially present in a treated locality satisfy $S_i(0) = 1$. This means that under control they would still be present at $t = 1$, so the pre-treatment

Figure 1: Principal strata in the no-turnover setting



Note: At $t = 0$, the treated area contains stayers and future escapees. At $t = 1$, it contains stayers and followers. Escapees leave and followers arrive. In the no-turnover setting, the control population is stable and both flows originate from or go to outside the study population.

treated population is $\mathcal{S} \cup \mathcal{L}$ and all exits from treated localities are treatment-induced. Let $\pi_S, \pi_L, \pi_A, \pi_N$ denote the shares of the four strata, with $\pi_S + \pi_L + \pi_A + \pi_N = 1$. The stayer share in the pre-treatment observed population is

$$w_0 = \frac{\pi_S}{\pi_S + \pi_L},$$

and the stayer share in the post-treatment observed population is

$$w_1 = \frac{\pi_S}{\pi_S + \pi_A}.$$

For simplicity of exposition, here the control group is assumed to remain at $t = 1$.

Assumption 1 (Stable Control Population)**.** *For all agents initially in control localities, $S_i(d) = 1$ for all $d \in \{0, 1\}$.*

Because the same control agents are observed in both periods, the control mean change can be read as a genuine within-population trend rather than a mixture of trend and turnover.

This setting also assumes that treatment in treated areas does not affect control areas.

Assumption 2 (No Interference)**.** *For all agents initially in control localities, $Y_{it}(d, 1) = Y_{it}(0, 1)$ for all $d \in \{0, 1\}$.*

Under Assumptions 1 and 2, the control group is stable and untreated throughout. Together

Table 1: Observability of potential outcomes by principal stratum

Stratum	$d=1$			$d=0$
	$Y_{i0}(0,1)$ *baseline*	$Y_{i1}(1,1)$ *in locality*	$Y_{i1}(1,0)$ *at destination*	$Y_{i1}(0,1)$ *in locality*
$\mathcal{S}$: stayers	✓	✓	–	○
$\mathcal{L}$: escapees	✓	○†	✓*	○
$\mathcal{A}$: followers	–	✓	–	–
$\mathcal{N}$: never-present	–	–	–	–

Note: ✓ observed; ○ counterfactual (well-defined but not realized); ○† double counterfactual; ✓* realized but only if escapees are tracked to their destination; – agent absent from the locality under the stated (d, s).

with $S_i(0) = 1$ for initially present treated agents, these assumptions define the *no-turnover setting*: all composition changes are treatment-induced, and the control population serves as a clean counterfactual trend. This is of course a simplification, because both treated and control areas typically have natural turnover rates. Appendix A3 relaxes these restrictions and identifies conditions under which the no-turnover results carry through.

Followers therefore arrive from outside the study population rather than from control localities. Figure 1 summarizes the principal strata in this setting, where controls are unaffected by treatment and followers come from a third area. This scenario is often not realistic in cases of policy, but can be approximated by selecting control group locations that are more distant, or otherwise less prone to migration (Butts 2024).

Finally, as in standard DiD practice, I assume that agents do not respond in anticipation of the treatment.

Assumption 3 (No Anticipation)**.** *For all agents initially present in treated localities,* $Y_{i0}(1,1) = Y_{i0}(0,1)$.

No anticipation ensures that pre-treatment outcomes can be interpreted as clean baselines rather than early responses to expected treatment. Table 1 records which potential outcomes are observed and which remain counterfactual.

Causal quantities

What quantities do researchers target when evaluating policies with DiD designs? If the research question is about what happens to the locality as a locality, a potential quantity of

interest is the locality-level treatment effect

$$\tau^{LOC} \equiv \mathbb{E}\big[\bar{Y}_{j,1} - \bar{Y}_{j,1}(0) \mid j \in \mathcal{T}\big]. \tag{1}$$

This is a locality-level effect because it compares the expected post-treatment mean in treated localities with the mean that would have prevailed under control. The important point under compositional change is that the two means are not taken over the same agents. Under treatment the post-treatment population is $\mathcal{S} \cup \mathcal{A}$, whereas under control it would be $\mathcal{S} \cup \mathcal{L}$. For that reason, τ^{LOC} need not be an average of individual treatment effects over a fixed population. In order to better characterize τ^{LOC}, let us define other individual causal quantities.

If the question of interest is about the people who were exposed before treatment, the relevant population is the pre-treatment treated population $\mathcal{S} \cup \mathcal{L}$. Within that population, the Survivor Average Treatment Effect is

$$\tau^{SATE} \equiv \mathbb{E}\big[Y_{i1}(1,1) - Y_{i1}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}\big]. \tag{2}$$

This is the average treatment effect for stayers, holding location fixed at the treated locality. This might be a quantity of direct interest for policy evaluation even if deviating from the exposed-population ATT. For escapees, the analogous object is

$$\tau^{EATE} \equiv \mathbb{E}\big[Y_{i1}(1,0) - Y_{i1}(0,1) \mid i \in \mathcal{L},\, j \in \mathcal{T}\big]. \tag{3}$$

The first term is the escapee's outcome at the destination under treatment. The second is the outcome the same person would have had in the treated locality under control. The EATE is therefore a total effect that bundles the treatment response with the relocation that treatment induces.

Next, the ATT for the pre-treatment treated population is

$$\tau^{ATT} \equiv \mathbb{E}\big[Y_{i1}(1, S_i(1)) - Y_{i1}(0, S_i(0)) \mid i \in \mathcal{S} \cup \mathcal{L},\, j \in \mathcal{T}\big], \tag{4}$$

that is, the average difference in post-treatment potential outcomes of stayers and leavers under treatment and under control. In the simplified setting of no turnover, this can be written as

$$\tau^{ATT} = w_0\, \tau^{SATE} + (1 - w_0)\, \tau^{EATE}. \tag{5}$$

It is convenient to summarize the difference in treatment effects between escapees and stayers

with the selection parameter

$$\delta \equiv \tau^{EATE} - \tau^{SATE},$$

so that

$$\tau^{ATT} = \tau^{SATE} + (1 - w_0)\,\delta. \tag{6}$$

When $\delta = 0$, the ATT and the SATE coincide. When $\delta \neq 0$, the ATT differs from the stayer effect because escapees experience a different total causal effect from stayers. Meanwhile, the locality-level effect and the ATT are generally distinct even when $\delta = 0$. Expanding τ^{LOC} in terms of stratum means (see Appendix A2 for derivation) gives

$$\tau^{LOC} = \underbrace{w_1\,\tau^{SATE}}_{\text{weighted SATE}} + \underbrace{(w_1 - w_0)\,\bar{Y}_{\mathcal{S},1}(0,1)}_{\text{stayer reweighting}} + \underbrace{(1 - w_1)\,\bar{Y}_{\mathcal{A},1}(1,1) - (1 - w_0)\,\bar{Y}_{\mathcal{L},1}(0,1)}_{\text{entry and exit}}. \tag{7}$$

The expasion reflects three differences between locality and individual treatement effects. The locality-level effect weights stayers by the post-treatment share rather than the pre-treatment share, it includes followers, and it includes the escapees' counterfactual in-locality outcome under control rather than their realized destination outcome under treatment. These differences make τ^{LOC} difficult to interpret as a causal effect with direct ties to individual outcomes.

As shown in Appendix A1, the estimand distinctions collapse when treatment does not induce migration. If $\pi_L = \pi_A = 0$, then $w_0 = w_1 = 1$ and $\tau^{LOC} = \tau^{SATE} = \tau^{ATT}$. The framework above makes explicit what breaks once migration is part of the treatment response.

The choice of target estimand ultimately depends on the research question. If the object of interest is what happens to the locality, such as whether average wages in a city rise, then τ^{LOC} is the right target. If the object of interest is the outcomes of people who lived in treated areas before treatment, the right target is τ^{ATT}. If the object of interest is the direct behavioral response to treatment holding location fixed, researchers should target τ^{SATE}. The next section asks when the aggregate DiD estimand θ^{agg} and the within-individual DiD estimand θ^{ind} recover those causal quantities.

What the DiD Estimators Identify

DiD identifies causal effects by subtracting an observed control trend from an observed treated trend. The subtraction removes the untreated component of the treated trend only if that counterfactual component equals the observed control trend, and this is the content of a parallel trends assumption. Without migration, a single such assumption suffices to

identify all estimands. Once treatment changes local compositions, the locality-level and the individual-level estimands represent a different causal quantity, and hence require different versions of parallel trends.

Aggregate DiD and the locality-level effect

Let the aggregate DiD estimand be

$$\theta^{agg} \equiv \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{T}] - \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{C}]. \tag{8}$$

To understand what θ^{agg} identifies, write the expected counterfactual post-treatment mean in treated localities under control as

$$\begin{aligned}\mathbb{E}\big[\bar{Y}_{j,1}(0) \mid j \in \mathcal{T}\big] = w_0\, &\mathbb{E}[Y_{i1}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}] \\ &+ (1 - w_0)\, \mathbb{E}[Y_{i1}(0,1) \mid i \in \mathcal{L},\, j \in \mathcal{T}]. \end{aligned} \tag{9}$$

Under control, escapees do not leave in the no-turnover setting, so both stayers and would-be escapees remain in the locality. Let

$$\Delta_S^0 \equiv \mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}],$$

$$\Delta_L^0 \equiv \mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid i \in \mathcal{L},\, j \in \mathcal{T}],$$

and let

$$\Delta_C \equiv \mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid j \in \mathcal{C}]$$

denote the control trend. Because the control population is stable, Δ_C is simply the observed control mean change.

Assumption 4 (Aggregate Parallel Trends)**.**

$$\mathbb{E}[\bar{Y}_{j,1}(0) - \bar{Y}_{j,0}(0) \mid j \in \mathcal{T}] = \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{C}].$$

Without natural turnover, Assumption 4 is equivalent to

$$w_0\, \Delta_S^0 + (1 - w_0)\, \Delta_L^0 = \Delta_C. \tag{10}$$

Aggregate parallel trends therefore restricts a weighted average of the treated stayer and escapee counterfactual trends, not each subgroup trend individually. It can hold even if

stayers and escapees follow different untreated trajectories, provided the differences wash out in the aggregate.

Proposition 1 (Aggregate DiD and the locality-level effect)**.** *Under Assumptions 1, 2, and 4,*

$$\theta^{agg} = \tau^{LOC}.$$

Under aggregate parallel trends, the control comparison removes the common untreated trend, leaving the locality-level treatment effect (proof in Appendix A2). Aggre gate parallel trends is a locality-level balance condition rather than an individual-level behavioral law. It disciplines the locality mean, and that is all the locality-level question requires.

Within-individual DiD and the stayer effect

Let the within-individual DiD estimand be the coefficient θ^{ind} from the regression

$$Y_{ijt} = \rho_i + \gamma_t + \theta^{ind} D_{jt} + \varepsilon_{ijt},$$

estimated on individuals observed in both periods. Because within-individual DiD restricts attention to agents present in both periods, it holds composition fixed by design. To identify the stayer effect, it requires a parallel trends condition at the individual level.

Assumption 5 (Stayer Parallel Trends)**.**

$$\mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}] = \mathbb{E}[Y_{i1} - Y_{i0} \mid j \in \mathcal{C}].$$

Stayer parallel trends says that treated stayers and the stable control population would have followed the same trend absent treatment. In the shorthand above, it is simply $\Delta_S^0 = \Delta_C$. The right-hand side uses observed outcomes rather than potential outcomes because the control group is untreated and stable, so that $\mathbb{E}[Y_{i1} - Y_{i0} \mid j \in \mathcal{C}] = \mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid j \in \mathcal{C}]$. This is the closest analogue to the standard individual-level DiD intuition. The assumption permits selection on outcome levels, so that stayers may have systematically higher or lower outcomes than the control group, but it rules out selection on untreated outcome growth.

To see how the aggregate estimand separates into a behavioral component and a compositional component, write the treated first-difference as a stayer-change term and two compo-

sition terms reflecting selective exit and entry:

$$\begin{aligned}\mathbb{E}\big[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{T}\big] = &\underbrace{\big(\bar{Y}_{\mathcal{S},1}(1,1) - \bar{Y}_{\mathcal{S},0}(0,1)\big)}_{\text{stayer change}} \\ &+ \underbrace{(1-w_0)\big(\bar{Y}_{\mathcal{S},0}(0,1) - \bar{Y}_{\mathcal{L},0}(0,1)\big)}_{\text{exit composition}} \\ &+ \underbrace{(1-w_1)\big(\bar{Y}_{\mathcal{A},1}(1,1) - \bar{Y}_{\mathcal{S},1}(1,1)\big)}_{\text{entry composition}}. \end{aligned} \tag{11}$$

The last two terms are the compositional terms created by selective exit and selective entry. Denote their sum by

$$\theta^{comp} \equiv (1-w_0)\big(\bar{Y}_{\mathcal{S},0}(0,1) - \bar{Y}_{\mathcal{L},0}(0,1)\big) + (1-w_1)\big(\bar{Y}_{\mathcal{A},1}(1,1) - \bar{Y}_{\mathcal{S},1}(1,1)\big). \tag{12}$$

Proposition 2 (Aggregate DiD decomposition without natural turnover)**.** *Under Assumptions 1, 2, 3, and 5,*

$$\theta^{agg} = \tau^{SATE} + \theta^{comp}.$$

Proposition 2 shows that, without natural turnover, aggregate DiD combines a behavioral term and a compositional term. When Assumption 4 also holds, Propositions 1 and 2 together imply $\tau^{LOC} = \tau^{SATE} + \theta^{comp}$. This is consistent with the expansion of τ^{LOC} in Equation 7: the stayer-reweighting and entry/exit terms there absorb θ^{comp} together with the weight difference $(w_1 - 1)\,\tau^{SATE}$ that arises because Equation 7 scales the stayer effect by w_1 rather than by one.

Proposition 3 (Within-individual DiD)**.** *Under Assumptions 1, 2, 3, and 5,*

$$\theta^{ind} = \tau^{SATE}.$$

By restricting attention to agents observed in both periods, within-individual DiD removes the entry and exit terms from (11). What remains is the stayer change, and stayer parallel trends removes its untreated component.

Combining Propositions 2 and 3 yields the exact no-turnover identity

$$\theta^{agg} - \theta^{ind} = \theta^{comp}. \tag{13}$$

This diagnostic difference is useful because the gap between the aggregate place-based response and the stayer-based response it is exactly the compositional term without natural

turnover. Once natural turnover is restored, however, the same gap remains informative but its interpretation changes.

With both assumptions in hand, it is useful to note a joint implication. Combining (10) with stayer parallel trends yields

$$\Delta_L^0 = \Delta_C \qquad \text{if } w_0 < 1 \tag{14}$$

(proof in Appendix A2). Without natural turnover, the two parallel trends assumptions together force would-be escapees to share the control trend as well. Once combined with stayer parallel trends, aggregate parallel trends does not merely discipline the locality mean; it also constrains the untreated trajectory of a group selected precisely because treatment causes them to leave. The joint content of the two assumptions is therefore stronger than either one suggests on its own. Once natural turnover is restored, the expanded aggregate condition involves additional subgroup trends and the substitution argument no longer isolates the escapee trajectory. Appendix A3 shows that the corollary is recovered under the conditions that restore the full no-turnover decomposition.

Both assumptions are most plausible when migration is driven by level differences, meaning that agents leave because their outcome level is low relative to outside options, rather than by trend differences, meaning that agents leave because their outcomes are deteriorating. Selection on levels is compatible with parallel trends; selection on trends is not. The restrictions also become harder to sustain as the escapee share grows, because the escapee trend carries more weight in the aggregate condition. Under the extension with natural turnover discussed in Appendix A3, aggregate parallel trends becomes more composite, because it implicitly bundles subgroup trends and turnover margins in both treated and control areas.

The remaining distance from the stayer effect to the ATT is governed by the selection parameter:

$$\tau^{ATT} = \theta^{ind} + (1 - w_0)\,\delta. \tag{15}$$

Neither estimator identifies τ^{ATT} without information or assumptions about δ. Even with rich panel data, the counterfactual $Y_{i1}(0,1)$ for escapees is never observed.

Identification by data regime

The identification picture depends on what kind of individual information is available. Table 2 summarizes the no-turnover case.

With aggregate data alone, the only object that is point-identified without natural turnover is the locality-level effect. This is already enough if the question is genuinely place-based. It

Table 2: Identification status without natural turnover

	Data regime		
Estimand	Aggregate data only	Repeated cross-sections	Individual panel
τ^{LOC}	Point ID	Point ID	Point ID
τ^{SATE}	–	Bounded in one-sided exit case	Point ID
τ^{EATE}	–	–	Point ID†
τ^{ATT}	–	Bounded given a range for δ	Sensitivity region
θ^{comp}	–	–	Point ID‡

Notation: – Not identified; † If destinations are tracked; ‡ Without natural turnover.
Note: The table refers to the no-turnover setting with treatment-induced migration in treated areas, a stable control population, and no interference. The repeated cross-section entry is the one-sided case in which treatment induces exits but not entries.

is not enough if the objective is to make claims about the effects on stayers or pre-treatment residents.

Individual panel data

With individual panel data that track the same agents in both periods through persistent identifiers stayers, escapees, and followers can all be classified directly. Without natural turnover, within-individual DiD on panel data point-identify τ^{SATE}. Moreover, they also identify the compositional term through the diagnostic gap,

$$\theta^{comp} = \theta^{agg} - \theta^{ind},$$

or directly from the decomposition

$$\hat{\theta}^{comp} = (1 - \hat{w}_0)\,\hat{\Delta}_{SL} + (1 - \hat{w}_1)\,\hat{\Delta}_{AS}, \tag{16}$$

where

$$\hat{\Delta}_{SL} = \bar{Y}_{\mathcal{S},0}(0,1) - \bar{Y}_{\mathcal{L},0}(0,1)$$

is the pre-treatment stayer–escapee gap, and

$$\hat{\Delta}_{AS} = \bar{Y}_{\mathcal{A},1}(1,1) - \bar{Y}_{\mathcal{S},1}(1,1)$$

is the post-treatment follower–stayer gap.

Instead, identification of τ^{EATE} requires an additional data condition. When destinations are not tracked, τ^{EATE} is never identified. If the panel tracks escapees to their post-treatment

destinations, however, then $Y_{i1}(1,0)$ is observed for escapees. Without natural turnover, Assumptions 4 and 5 imply (14), so the escapee counterfactual trend equals the control trend. This yields

$$\hat{\tau}^{EATE} = \left[\bar{Y}_{\mathcal{L},1}(1,0) - \bar{Y}_{\mathcal{L},0}(0,1)\right] - \Delta_C. \tag{17}$$

The ATT can only be bounded within a sensitivity region unless the escapee effect is pinned down. If destinations are tracked, (17) can be used directly and the sensitivity exercise collapses to a point estimate. Otherwise, using (15) under Assumption 5 alone,

$$\tau^{ATT}(\delta) = \hat{\tau}^{SATE} + (1 - \hat{w}_0)\,\delta. \tag{18}$$

A simple way to calibrate δ to plausible differences in treatment effects is to use the observable pre-treatment stayer–escapee gap. One can report $\tau^{ATT}(\delta)$ for values satisfying $|\delta| \leq \kappa|\hat{\Delta}_{SL}|$, where κ indicates multiples of the gap such as $\{0.5, 1, 2\}$, together with the sign-reversal threshold

$$\delta^* = -\frac{\hat{\tau}^{SATE}}{1 - \hat{w}_0}.$$

This sensitivity exercise describes the sign and magnitude of τ^{ATT} for empirically-grounded values of δ.

Repeated cross-sections

Repeated cross-sections are less informative than true panels because individuals are observed in each period but cannot be linked across periods. Because of this information problem, causal quantities can only be partially identified in the one-sided case where treatment induces exits and not entries.

Assumption 6 (Monotone Exit Migration)**.** *Treatment weakly increases the probability of leaving:* $S_i(1) \leq S_i(0)$ *for all* i*.*

Under Assumption 6, there are no followers. This is a weak assumption under no turnover, as control exit is already restricted. The post-treatment treated sample then consists of stayers, while the pre-treatment treated sample mixes stayers and future escapees. Let

$$\hat{q}_L = \hat{P}(\text{leave} \mid \mathcal{T}) - \hat{P}(\text{leave} \mid \mathcal{C})$$

denote the excess exit rate, where $\hat{P}(\text{leave} \mid \mathcal{C}) = 0$ under no turnover. A Lee-type bound (Lee 2009) trims a fraction $\hat{q}_L$ from the tails of the pre-treatment treated outcome distribution and uses the resulting trimmed means to bracket the unobserved pre-treatment mean of stayers.

Running DiD with those trimmed pre-treatment means yields lower and upper bounds for τ^{SATE}. The derivation is reported in Appendix A4.

The same logic can be carried over to the ATT through Equation 18. If the SATE is bounded and δ is restricted to a plausible range $[\delta_{\min}, \delta_{\max}]$, then the ATT is bounded as well. What repeated cross-sections do not provide is a pre-treatment stayer–escapee gap useful for calibrating δ. That calibration must come from theory, outside evidence, or auxiliary data such as population-representative surveys (Marbach 2025).

When treatment induces both exits and entries, a similar trimming approach would require additional structure on how entry and exit margins behave that is beyond the scope of this paper. The point here is that repeated cross-sections can be useful, but under selective migration they support only a narrower and more assumption-dependent analysis than individual panels. Appendix A3 shows that once untreated turnover and treatment-induced retention are admitted, the one-sided trimming result does not extend mechanically, and repeated cross-sections support place-based analysis more naturally than person-based analysis unless additional structure is imposed.

Discussion

When treatment changes who is present in a locality, aggregate DiD and within-individual DiD no longer answer the same question. The aggregate estimator speaks to what happened to the locality. The within-individual estimator speaks to what happened to stayers. The ATT for the pre-treatment population sits between those objects and requires additional information about escapees to be identified.

The no-turnover setting presented in this paper is deliberately simple. It assumes that composition changes come from treatment-induced migration in treated areas, while the control population is stable. That simplification is useful because it isolates the estimand logic cleanly. Without natural turnover, aggregate DiD identifies the locality-level effect under aggregate parallel trends, within-individual DiD identifies the SATE under stayer parallel trends, and the gap between the two estimators exactly measures the compositional term.

The extensions in the appendix show why the simplification matters. Once natural turnover is restored, control-group composition enters the aggregate comparison and the set of agents observed in both periods under treatment need not coincide with the always-stayers of the no-turnover case. The interpretation of the aggregate DiD becomes a composite of more strata, and the same is true for the within-individual DiD.

With natural turnover, panel data still identify the effect for the treated agents observed in both periods, but this is no longer the stayer effect of the no-turnover setting. The observed panel sample pools always-stayers with *retained* agents, that is individuals whose presence at $t = 1$ is itself a treatment response because they would have left under control. Because the two groups cannot be separated using realized treated-panel data alone, the SATE for always-stayers is not point-identified without assumptions that rule out treatment-induced retention, such as monotone exit migration. The escapee effect is likewise not point-identified, because actual leavers under treatment pool treatment-induced leavers with agents who would have left regardless. The clean one-dimensional sensitivity for τ^{ATT} no longer applies.

For repeated cross-sections, the Lee-bounds argument depends on monotonicity, which simultaneously rules out retained agents and enables trimming. Once treatment-induced retention is admitted, the post-treatment treated sample is no longer composed of pure stayers, and the excess exit rate no longer cleanly estimates the treatment-induced exit share. Repeated cross-sections then support bounds only under additional restrictions, or place-based analysis via aggregate DiD.

The no-turnover results do carry through, however, under two necessary conditions. A first treated-side condition requires that all initially present agents would have remained under contro. This rules out both retained agents and natural leavers and ensures that within-individual DiD identifies the always-stayer effect. A common-natural-turnover condition instead imposes that the composition effect from treatment-independent exits and entries is the same in treated and control areas, and therefore cancels in the DiD comparison. These conditions are plausible when treated and control areas are selected to have similar background mobility patterns, which is already implied in many DiD designs.

The take away of this paper is that researchers should be mindful of which causal quantity they need to anser a research question (Slough 2023). If the question is about the locality, aggregate DiD may already be the right estimand. If the question is about pre-treatment residents, the aggregate estimate and the within-individual estimate should not be treated as interchangeable. The estimand problem should come first, and the data regime determines how far one can go after that.

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Supplementary Materials

A1 DiD with a fixed population

This appendix states the standard DiD result for a setting without migration. The purpose is to show that the estimand distinctions developed in the main text do not arise when the population is fixed, and to make explicit what migration breaks.

Consider agents $i \in \{1, \ldots, N_j\}$ in locality j, observed over two periods $t \in \{0, 1\}$. Treatment $D_j \in \{0, 1\}$ is assigned at the locality level and activated at $t = 1$. The same agents are present in both periods. Because no agent moves, potential outcomes depend only on the treatment status of the locality, not on whether the agent remains there. Write them as $Y_{it}(d)$ for treatment status $d \in \{0, 1\}$.

The locality-level treatment effect is the change in the locality post-treatment mean caused by treatment. Because the population is fixed, both the treated and counterfactual means are taken over the same agents:

$$\tau^{LOC} = \bar{Y}_{j,1}(1) - \bar{Y}_{j,1}(0) = \frac{1}{N_j} \sum_{i=1}^{N_j} \big[Y_{i1}(1) - Y_{i1}(0)\big]. \tag{A.1}$$

The individual ATT is the expected treatment effect for agents in treated localities:

$$\tau^{ATT} = \mathbb{E}\big[Y_{i1}(1) - Y_{i1}(0) \mid j \in \mathcal{T}\big].$$

With a fixed population, $\tau^{LOC} = \tau^{ATT}$. The locality-level effect is an average of individual treatment effects over the same agents who define the ATT. There is no compositional distinction. Now consider identification under no anticipation, so that $Y_{i0}(1) = Y_{i0}(0)$, and standard parallel trends:

$$\mathbb{E}\big[Y_{i1}(0) - Y_{i0}(0) \mid j \in \mathcal{T}\big] = \mathbb{E}\big[Y_{i1}(0) - Y_{i0}(0) \mid j \in \mathcal{C}\big].$$

The aggregate DiD estimand is

$$\theta^{agg} = \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{T}] - \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{C}].$$

With a fixed population, the locality mean change $\bar{Y}_{j,1} - \bar{Y}_{j,0}$ is the average of individual first-differences, so θ^{agg} equals the individual-level DiD estimand θ^{ind} obtained from the within-individual regression with agent fixed effects. Under no anticipation and parallel trends, both equal τ^{ATT}:

$$\theta^{agg} = \theta^{ind} = \tau^{ATT} = \tau^{LOC}.$$

All four objects coincide. The aggregate and individual estimators are interchangeable, and the locality-level question and the individual-level question have the same answer. Once migration is allowed, however, the population observed at $t = 1$ differs from the one at $t = 0$. The potential-outcome notation expands from $Y_{it}(d)$ to $Y_{it}(d, s)$ to track whether the agent is present, and the equivalences above no longer hold. The main text develops the framework needed to study what each estimator identifies in that setting.

A2 Proofs

Derivation of Equation 7

Start from the definition of the locality-level treatment effect in Equation 1. Under treatment, the post-treatment population is $\mathcal{S} \cup \mathcal{A}$ with stayer weight w_1. Under control, it would be $\mathcal{S} \cup \mathcal{L}$ with stayer weight w_0. The two expected means are therefore

$$\begin{aligned}
\mathbb{E}\big[\bar{Y}_{j,1} \mid j \in \mathcal{T}\big] &= w_1\, \bar{Y}_{\mathcal{S},1}(1,1) + (1 - w_1)\, \bar{Y}_{\mathcal{A},1}(1,1), \\
\mathbb{E}\big[\bar{Y}_{j,1}(0) \mid j \in \mathcal{T}\big] &= w_0\, \bar{Y}_{\mathcal{S},1}(0,1) + (1 - w_0)\, \bar{Y}_{\mathcal{L},1}(0,1).
\end{aligned}$$

Subtracting and adding and subtracting $w_1\, \bar{Y}_{\mathcal{S},1}(0,1)$ to isolate $\tau^{SATE} = \bar{Y}_{\mathcal{S},1}(1,1) - \bar{Y}_{\mathcal{S},1}(0,1)$ gives

$$\begin{aligned}
\tau^{LOC} &= w_1 \big[\bar{Y}_{\mathcal{S},1}(1,1) - \bar{Y}_{\mathcal{S},1}(0,1)\big] + (w_1 - w_0)\, \bar{Y}_{\mathcal{S},1}(0,1) \\
&\quad + (1 - w_1)\, \bar{Y}_{\mathcal{A},1}(1,1) - (1 - w_0)\, \bar{Y}_{\mathcal{L},1}(0,1) \\
&= w_1\, \tau^{SATE} + (w_1 - w_0)\, \bar{Y}_{\mathcal{S},1}(0,1) + (1 - w_1)\, \bar{Y}_{\mathcal{A},1}(1,1) - (1 - w_0)\, \bar{Y}_{\mathcal{L},1}(0,1).
\end{aligned}$$

□

Proof of Proposition 1

By definition,

$$\theta^{agg} = \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{T}] - \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{C}].$$

Under Assumption 2, the control group is untreated, and under Assumption 1 it is the same population in both periods. Assumption 4 therefore implies that the control first-difference equals the untreated trend that would have prevailed in treated localities under $D = 0$. Subtracting it from the treated first-difference yields

$$\theta^{agg} = \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,1}(0) \mid j \in \mathcal{T}] = \tau^{LOC}.$$

□

Proof of Equation 14

Substitute $\Delta_S^0 = \Delta_C$ (Assumption 5) into the expanded form of aggregate parallel trends in (10):

$$w_0\, \Delta_C + (1 - w_0)\, \Delta_L^0 = \Delta_C.$$

Rearranging gives $(1 - w_0)\, \Delta_L^0 = (1 - w_0)\, \Delta_C$. Since $w_0 < 1$, dividing both sides by $(1 - w_0)$ yields $\Delta_L^0 = \Delta_C$. □

Proof of Proposition 2

Start from the treated first-difference decomposition in (11). By Assumption 3,

$$\bar{Y}_{\mathcal{S},0}(0,1) = \bar{Y}_{\mathcal{S},0}(1,1),$$

so the stayer-change term can be written as

$$\bar{Y}_{\mathcal{S},1}(1,1) - \bar{Y}_{\mathcal{S},0}(0,1) = \tau^{SATE} + \Delta_S^0.$$

Under Assumption 5, $\Delta_S^0 = \Delta_C$. Because the control population is stable, the control first-difference equals Δ_C. Subtracting the control first-difference from the treated first-difference therefore gives

$$\theta^{agg} = \tau^{SATE} + \theta^{comp}.$$

□

Proof of Proposition 3

Without natural turnover, the within-individual sample in treated localities consists of stayers only. For such agents,

$$Y_{i1} - Y_{i0} = Y_{i1}(1,1) - Y_{i0}(0,1).$$

The within-individual DiD compares the mean of this first-difference in treated stayers with the observed first-difference in the stable control population. Taking expectations and using Assumption 3,

$$\begin{aligned}\theta^{ind} &= \mathbb{E}[Y_{i1}(1,1) - Y_{i0}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}] - \mathbb{E}[Y_{i1} - Y_{i0} \mid j \in \mathcal{C}] \\ &= \tau^{SATE} + \Delta_S^0 - \Delta_C.\end{aligned}$$

Assumption 5 gives $\Delta_S^0 = \Delta_C$, so $\theta^{ind} = \tau^{SATE}$. □

A3 Extension: Allowing natural turnover

The main text imposed two simplifying restrictions. First, among agents initially present in treated localities, everyone would have remained under control. Second, the control group was stable across the two periods. This appendix relaxes both restrictions. The point is not to build a different paper, but to show what changes once untreated turnover is allowed back in.

Among agents initially present in a treated locality, four strata now arise:

- $\mathcal{S}$: present under both control and treatment,
- $\mathcal{L}$: present under control and absent under treatment,
- $\mathcal{R}$: absent under control and present under treatment,
- $\mathcal{N}$: absent under both control and treatment.

Relative to the main text, the new stratum is $\mathcal{R}$, which I call *retained* agents. These are agents whose presence at $t = 1$ is itself treatment-induced retention. They would have left under control, but they remain under treatment.

Meanwhile, among agents initially absent from the treated locality define:

- $\mathcal{B}$: present under both control and treatment,
- $\mathcal{A}$: present only under treatment,
- $\mathcal{Q}$: present only under control,
- $\mathcal{M}$: absent under both control and treatment.

Here $\mathcal{B}$ are natural entrants, $\mathcal{A}$ are followers, and $\mathcal{Q}$ are entrants crowded out by treatment.

The post-treatment population in a treated locality is now

$$\mathcal{P}_1(1) = \mathcal{S} \cup \mathcal{R} \cup \mathcal{B} \cup \mathcal{A}$$

under treatment, and

$$\mathcal{P}_1(0) = \mathcal{S} \cup \mathcal{L} \cup \mathcal{B} \cup \mathcal{Q}$$

under control. The locality-level effect therefore becomes

$$\tau^{LOC,gen} \equiv \mathbb{E}\big[\bar{Y}_{j,1} - \bar{Y}_{j,1}(0) \mid j \in \mathcal{T}\big]. \tag{A.2}$$

Compared with the main text, the counterfactual and observed post-treatment means differ along more than one composition margin.

For the actual treated first-difference, let

$$\mathcal{SR} \equiv \mathcal{S} \cup \mathcal{R}, \qquad \mathcal{LN} \equiv \mathcal{L} \cup \mathcal{N},$$

and define

$$v_0 \equiv \frac{N_S + N_R}{N_S + N_R + N_L + N_N}, \qquad u_B \equiv \frac{N_B}{N_S + N_R + N_B + N_A}, \qquad u_A \equiv \frac{N_A}{N_S + N_R + N_B + N_A}.$$

Then

$$\begin{aligned}
\bar{Y}_{T,1} - \bar{Y}_{T,0} = {} & \big(\bar{Y}_{\mathcal{SR},1}(1,1) - \bar{Y}_{\mathcal{SR},0}(0,1)\big) \\
& + (1 - v_0)\big(\bar{Y}_{\mathcal{SR},0}(0,1) - \bar{Y}_{\mathcal{LN},0}(0,1)\big) \\
& + u_B\big(\bar{Y}_{\mathcal{B},1}(1,1) - \bar{Y}_{\mathcal{SR},1}(1,1)\big) \\
& + u_A\big(\bar{Y}_{\mathcal{A},1}(1,1) - \bar{Y}_{\mathcal{SR},1}(1,1)\big).
\end{aligned} \tag{A.3}$$

The difference from the main text is that the pre-treatment composition term now pools treatment-induced leavers and natural leavers, and the post-treatment composition term now contains both natural entrants and treatment-induced followers.

Control areas also regain their own turnover terms. Let $\mathcal{S}^C$ denote control stayers, $\mathcal{L}^C$ control leavers, and $\mathcal{A}^C$ control arrivals. Then

$$\bar{Y}_{C,1} - \bar{Y}_{C,0} = \underbrace{(\bar{Y}_{\mathcal{S}^C,1} - \bar{Y}_{\mathcal{S}^C,0})}_{\Delta_C} + \underbrace{(1 - w_0^C)(\bar{Y}_{\mathcal{S}^C,0} - \bar{Y}_{\mathcal{L}^C,0}) + (1 - w_1^C)(\bar{Y}_{\mathcal{A}^C,1} - \bar{Y}_{\mathcal{S}^C,1})}_{\theta^{comp,C}}. \tag{A.4}$$

So the aggregate comparison regains a control-side composition term as well.

The interpretation of within-individual DiD changes at the same time. In treated localities, the agents observed in both periods are now $\mathcal{S} \cup \mathcal{R}$ rather than $\mathcal{S}$ alone. The corresponding behavioral estimand is

$$\tau^{OBS} \equiv \mathbb{E}\big[Y_{i1}(1,1) - Y_{i1}(0,1) \mid i \in \mathcal{S} \cup \mathcal{R},\, j \in \mathcal{T}\big]. \tag{A.5}$$

Under a parallel-trends condition for this observed-stayer group,

$$\theta^{ind} = \tau^{OBS}.$$

The within-individual DiD therefore follows the agents who are observed in both periods under treatment. That is not, in general, the same group as the always-stayers $\mathcal{S}$ from the main text, and τ^{OBS} is not a relabeled SATE. The main-text τ^{SATE} is defined for always-stayers, agents whose location is treatment-invariant. The $\mathcal{R}$ stratum, is composed of individuals whose presence at $t = 1$ is itself a treatment response. They would have left under control, but remain under treatment. Including them changes the estimand from an always-stayer effect to an observed-stayer-under-treatment effect.

The interpretive difficulty is symmetric to the one flagged for escapees in Table 1. For escapees, the potential outcome $Y_{i1}(1,1)$ is a double counterfactual: it asks what the escapee's outcome would have been in the treated locality under treatment, even though the escapee is not there. For retained agents, $Y_{i1}(0,1)$ poses the same problem in the opposite direction: it asks what the retained agent's outcome would have been in the treated locality under control, even though under control the agent would have left. Both objects are well-defined in the potential-outcomes framework, but neither is empirically grounded in observable behavior.

Monotone exit migration (Assumption 6) rules out $\mathcal{R}$ entirely, so under monotonicity $\tau^{OBS} = \tau^{SATE}$. This is not incidental. Monotonicity does triple duty in the paper: it eliminates the retained stratum $\mathcal{R}$ that contaminates the within-individual estimand, it ensures that within-individual DiD identifies the always-stayer effect rather than a broader observed-stayer effect, and it enables Lee-type trimming for repeated cross-sections.

With untreated turnover, the aggregate decomposition is naturally written as

$$\theta^{agg} = \tau^{OBS} + \tilde{\theta}^{comp,T} - \theta^{comp,C}, \tag{A.6}$$

where $\tilde{\theta}^{comp,T}$ is the sum of the three composition terms in (A.3). The diagnostic gap becomes

$$\theta^{agg} - \theta^{ind} = \tilde{\theta}^{comp,T} - \theta^{comp,C}.$$

This is still informative, but it is no longer the same object as the main-text θ^{comp}, and it is centered on τ^{OBS} rather than on τ^{SATE}.

The main-text identities are recovered as a special case. If $\mathcal{R}$, $\mathcal{N}$, $\mathcal{B}$, $\mathcal{Q}$, $\mathcal{L}^C$, and $\mathcal{A}^C$ are all empty, then $\tau^{OBS} = \tau^{SATE}$, Equation A.3 collapses to (11), and Equation A.6 collapses to the no-turnover

decomposition. The no-turnover setting is useful because it isolates the estimand logic cleanly, while this appendix makes clear exactly what that simplification is buying.

Identification under turnover requires parallel trends assumptions that generalize the main-text Assumptions 4 and 5. The formal statements are identical in structure, but their content is richer because both sides of the comparison now involve composition changes.

Assumption A.1 (Generalized Aggregate Parallel Trends)**.**

$$\mathbb{E}[\bar{Y}_{j,1}(0) - \bar{Y}_{j,0} \mid j \in \mathcal{T}] = \mathbb{E}[\bar{Y}_{j,1} - \bar{Y}_{j,0} \mid j \in \mathcal{C}].$$

The mathematical form is the same as Assumption 4, but the objects on each side are different. In the main text, the treated counterfactual trend involves only the stable population $\mathcal{S} \cup \mathcal{L}$, and the control trend involves a stable control group. Under turnover, the treated counterfactual population shifts from $\mathcal{S} \cup \mathcal{L} \cup \mathcal{R} \cup \mathcal{N}$ at $t = 0$ to $\mathcal{S} \cup \mathcal{L} \cup \mathcal{B} \cup \mathcal{Q}$ at $t = 1$, and the control population shifts from $\mathcal{S}^C \cup \mathcal{L}^C$ to $\mathcal{S}^C \cup \mathcal{A}^C$.

To see what the assumption requires, decompose the treated counterfactual first-difference by separating the outcome trend for agents present in both periods under control ($\mathcal{S} \cup \mathcal{L}$) from the composition changes due to natural exits ($\mathcal{R} \cup \mathcal{N}$ departing) and entries ($\mathcal{B} \cup \mathcal{Q}$ arriving). Let $p_0 = (N_S + N_L)/(N_S + N_L + N_R + N_N)$ denote the share of would-stay-under-control agents in the pre-treatment treated population, and let $q_0 = (N_B + N_Q)/(N_S + N_L + N_B + N_Q)$ denote the share of entrants in the counterfactual post-treatment population. The expanded form of Assumption A.1 is then

$$w_0\, \Delta_S^0 + (1 - w_0)\, \Delta_L^0 + (1 - p_0)\big(\bar{Y}_{\mathcal{SL},0} - \bar{Y}_{\mathcal{RN},0}\big) + q_0\big(\bar{Y}_{\mathcal{BQ},1}(0,1) - \bar{Y}_{\mathcal{SL},1}(0,1)\big) = \Delta_C + \theta^{comp,C}, \quad \text{(A.7)}$$

where $\bar{Y}_{\mathcal{SL},0}$ and $\bar{Y}_{\mathcal{RN},0}$ are the pre-treatment means of would-stay and would-leave agents, $\bar{Y}_{\mathcal{BQ},1}(0,1)$ and $\bar{Y}_{\mathcal{SL},1}(0,1)$ are the counterfactual post-treatment means of entrants and stayers under control, Δ_C is the control stayer trend, and $\theta^{comp,C}$ is the control-side composition term from Equation A.4. Compared with the main-text expanded form in Equation 10, the left-hand side gains two composition terms reflecting natural exits and entries, and the right-hand side gains control-side composition.

Assumption A.2 (Observed-Stayer Parallel Trends)**.**

$$\mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid i \in \mathcal{S} \cup \mathcal{R},\, j \in \mathcal{T}] = \mathbb{E}[Y_{i1} - Y_{i0} \mid i \in \mathcal{S}^C,\, j \in \mathcal{C}].$$

This is the turnover analogue of Assumption 5. It requires that agents observed in both periods under treatment ($\mathcal{S} \cup \mathcal{R}$) would have followed the same untreated trend as control stayers ($\mathcal{S}^C$). Under the no-turnover setting, $\mathcal{R}$ is empty and $\mathcal{S}^C$ is the entire control population, so Assumption A.2 reduces to Assumption 5. The right-hand side uses observed outcomes because control stayers are untreated and present in both periods, so that $Y_{it} = Y_{it}(0,1)$ for $i \in \mathcal{S}^C$.

Identification under turnover

What does the extension imply for the identification results of Section ? Table A1 summarizes the picture.

With aggregate data, $\tau^{LOC,gen}$ remains point-identified under Assumption A.1. The argument is the

Table A1: Identification status under the turnover extension

Estimand	Data regime		
	Aggregate data only	Repeated cross-sections	Individual panel
$\tau^{LOC,gen}$ (locality-level effect)	Point ID	Point ID	Point ID
τ^{OBS} (observed-stayer effect)	Not identified	Not identified	Point ID
τ^{SATE} (always-stayer effect)	Not identified	Not identified	Not identified
τ^{EATE} (escapee total effect)	Not identified	Not identified	Not identified
τ^{ATT} (pre-treatment population)	Not identified	Not identified	Not identified

Note: The table refers to the general turnover extension without additional restrictions. Point identification of $\tau^{LOC,gen}$ uses Assumption A.1 (generalized aggregate parallel trends). Point identification of τ^{OBS} uses Assumption A.2 (observed-stayer parallel trends). Entries marked "not identified" require either monotonicity or other structure to recover the no-turnover results.

same as for Proposition 1: the control comparison removes the common counterfactual trend, leaving $\theta^{agg} = \tau^{LOC,gen}$. The assumption is more composite than its main-text counterpart, however, because the expanded form (A.7) bundles subgroup trends and turnover margins on both sides rather than comparing a stable treated population to a stable control population. In particular, the escapee parallel trends corollary of Equation 14 no longer follows from the joint content of aggregate and observed-stayer parallel trends. In the no-turnover setting, the expanded aggregate condition (10) has two treated-side unknowns (Δ_S^0 and Δ_L^0), and stayer parallel trends pins the first, so the second is determined. Under general turnover, the expanded condition (A.7) involves additional composition terms on the left-hand side and control-side composition on the right-hand side. Assumption A.2 pins only the $\mathcal{S} \cup \mathcal{R}$ trend, leaving the remaining unknowns undetermined. One can no longer isolate Δ_L^0 from a single aggregate restriction.

With individual panel data, θ^{ind} point-identifies τ^{OBS} under Assumption A.2, by the same argument as for Proposition 3: the within-individual comparison removes the common counterfactual trend for agents observed in both periods, leaving $\theta^{ind} = \tau^{OBS}$. The diagnostic gap $\theta^{agg} - \theta^{ind} = \tilde{\theta}^{comp,T} - \theta^{comp,C}$ is computable if control-side composition can be estimated separately, but it now conflates treated-side and control-side composition rather than isolating treatment-induced composition alone.

The always-stayer effect τ^{SATE} is not point-identified because $\mathcal{S}$ and $\mathcal{R}$ cannot be separated in the realized treated-panel data. The researcher observes who stayed under the actual treatment assignment, but not who would have stayed under the counterfactual assignment. Similarly, τ^{EATE} is not point-identified because actual leavers under treatment pool treatment-induced leavers ($\mathcal{L}$) and agents who would have left regardless ($\mathcal{N}$). Even when destinations are tracked, the observed destination outcomes belong to $\mathcal{L} \cup \mathcal{N}$, and stratum membership cannot be inferred from realized outcomes alone.

The ATT for the pre-treatment population is likewise not identified in the general case. Without natural turnover, the pre-treatment population consists of $\mathcal{S} \cup \mathcal{L}$, and the ATT reduces to $\tau^{SATE} + (1 - w_0)\,\delta$ with a single selection parameter $\delta = \tau^{EATE} - \tau^{SATE}$. Once natural turnover is restored, the pre-treatment population includes $\mathcal{S}$, $\mathcal{L}$, $\mathcal{R}$, and $\mathcal{N}$, and the ATT becomes a richer mixture that cannot be reduced to a one-parameter sensitivity problem. The sensitivity exercise of Section is therefore specific to the no-turnover setting.

With repeated cross-sections, the Lee-style argument of Appendix A4 requires monotone exit migration (Assumption 6), which rules out $\mathcal{R}$ and makes $\tau^{OBS} = \tau^{SATE}$. Once treatment-induced retention

is admitted, the post-treatment treated sample is no longer composed of pure stayers, and the excess exit rate mixes treatment-induced and natural exits. Without additional structure, repeated cross-sections support place-based analysis through τ^{LOC} but not person-based analysis.

When the no-turnover results carry through

Two additional conditions restore the no-turnover identification results in the presence of natural turnover.

The first restricts the treated side. The no-turnover setting assumes that all agents initially present in a treated locality would remain there under control, so that $S_i(0) = 1$ for every initially present agent. This assumption rules out both $\mathcal{R}$ (retained agents with $S_i(0) = 0, S_i(1) = 1$) and $\mathcal{N}$ (natural leavers with $S_i(0) = 0, S_i(1) = 0$). With both strata empty, the within-individual sample in treated areas consists of always-stayers only, so $\tau^{OBS} = \tau^{SATE}$, and the pre-treatment treated population reduces to $\mathcal{S} \cup \mathcal{L}$. This is weaker than requiring no treated-side turnover at all, because it allows treatment-induced exits ($\mathcal{L}$) and treatment-induced entries ($\mathcal{A}$). It only requires that, absent treatment, no one would have left.

Note that monotone exit migration (Assumption 6) is weaker than this condition. Monotonicity requires $S_i(1) \leq S_i(0)$, which rules out $\mathcal{R}$ but not $\mathcal{N}$, because natural leavers satisfy $S_i(1) = 0 \leq S_i(0) = 0$. Under monotonicity alone, the within-individual sample is still pure $\mathcal{S}$ (since both $\mathcal{L}$ and $\mathcal{N}$ leave under treatment and $\mathcal{R}$ does not exist), so $\tau^{OBS} = \tau^{SATE}$. However, the pre-treatment population includes $\mathcal{N}$ as well as $\mathcal{S}$ and $\mathcal{L}$, and the ATT becomes a three-group mixture rather than the two-group decomposition of the main text. For the ATT formula $\tau^{ATT} = \tau^{SATE} + (1 - w_0)\,\delta$ to apply exactly, one needs the stronger condition $S_i(0) = 1$.

The second condition handles the control side.

Assumption A.3 (Common Natural Turnover)**.** *The composition effect from treatment-independent exits and entries is the same in treated and control areas:*

$$\mathbb{E}[\theta^{nat}(j) \mid j \in \mathcal{T}] = \mathbb{E}[\theta^{nat}(j) \mid j \in \mathcal{C}],$$

where $\theta^{nat}(j)$ denotes the composition effect from turnover that would occur regardless of treatment status.

This is a composition-level parallel assumption, analogous to how standard parallel trends assumes common outcome trends. It requires that the selection patterns of natural movers, meaning the gaps between stayers and natural leavers at baseline and between natural arrivals and stayers post-treatment, are the same in treated and control areas.

Under these two conditions together, the generalized parallel trends assumptions reduce to their main-text counterparts. Consider first Assumption A.2 (observed-stayer parallel trends). Because $S_i(0) = 1$ rules out $\mathcal{R}$, the treated observed-stayer group $\mathcal{S} \cup \mathcal{R}$ reduces to $\mathcal{S}$ alone. Assumption A.2 therefore becomes

$$\mathbb{E}[Y_{i1}(0,1) - Y_{i0}(0,1) \mid i \in \mathcal{S},\, j \in \mathcal{T}] = \mathbb{E}[Y_{i1} - Y_{i0} \mid i \in \mathcal{S}^C,\, j \in \mathcal{C}],$$

which is Assumption 5 (stayer parallel trends) with the control side restricted to control stayers $\mathcal{S}^C$. The within-individual DiD therefore identifies τ^{SATE} rather than τ^{OBS}.

Consider next Assumption A.1 (generalized aggregate parallel trends). Under $S_i(0) = 1$, the strata $\mathcal{R}$ and $\mathcal{N}$ are empty, so $p_0 = 1$ in the expanded form (A.7) and the natural-exit composition term vanishes. The remaining natural-entry composition on the left-hand side (from $\mathcal{B}$) and the control-side composition $\theta^{comp,C}$ on the right-hand side reflect treatment-independent turnover. Common natural turnover (Assumption A.3) ensures that these terms cancel, because the composition effects from natural movers are the same in treated and control areas. What remains is

$$w_0\,\Delta_S^0 + (1 - w_0)\,\Delta_L^0 = \Delta_C,$$

which is exactly the main-text expanded form in Equation 10. The aggregate DiD therefore identifies τ^{LOC} rather than the broader $\tau^{LOC,gen}$, the diagnostic gap $\theta^{agg} - \theta^{ind}$ measures treatment-induced composition only, and the one-dimensional ATT sensitivity in δ remains valid. The main-text Table 2 applies under these two conditions even in the presence of natural turnover.

The escapee parallel trends corollary of Equation 14 is also recovered, because the reduced expanded form has the same two-unknown structure as in the no-turnover setting. Stayer parallel trends pins $\Delta_S^0 = \Delta_C$, and the substitution argument from Appendix A2 goes through unchanged. The corollary also holds under the weaker combination of monotonicity and common natural turnover. Monotonicity eliminates $\mathcal{R}$, and the departure of $\mathcal{N}$ agents is a natural-turnover event that cancels under the common-turnover condition, so the expanded aggregate condition again reduces to the $\mathcal{S}$-and-$\mathcal{L}$ structure.

Neither condition alone is sufficient. Without $S_i(0) = 1$, the within-individual estimand may pool $\mathcal{S}$ and $\mathcal{R}$ (if monotonicity fails) or the ATT may involve natural leavers (even if monotonicity holds). Without common natural turnover, the diagnostic gap conflates treatment-induced and control-side composition. Both conditions are needed to recover the full no-turnover decomposition.

The common-natural-turnover condition is most plausible when treated and control areas are selected to have similar background mobility patterns, which is already standard practice in DiD research designs. It is less plausible when treatment is assigned to areas with structurally different turnover, for instance when treated areas are rapidly growing cities while controls are stable towns.

A4 Lee-type bounds in the one-sided case

This appendix derives the repeated cross-section result for the one-sided case in which treatment induces exits but not entries. Under Assumption 6, the post-treatment treated sample consists of stayers, whereas the pre-treatment treated sample consists of stayers and future escapees.

Let $\mu_{T,1}$ denote the post-treatment treated mean and let $\mu_{C,1} - \mu_{C,0}$ denote the control first-difference. The difficulty is the pre-treatment treated mean, because it mixes stayers and future escapees. Monotone exit migration identifies the fraction to trim, not the identity of the agents who leave. The excess exit rate

$$q_L = P(\text{leave} \mid \mathcal{T}) - P(\text{leave} \mid \mathcal{C})$$

is the share of the pre-treatment treated population that is missing at $t = 1$ because of treatment.

Let $\mu_{T,0}^{lo}$ be the trimmed pre-treatment treated mean obtained after removing a fraction q_L from the lower tail of the pre-treatment treated outcome distribution, and let $\mu_{T,0}^{hi}$ be the analogous mean obtained by trimming from the upper tail. The unknown pre-treatment mean of stayers must lie

between these two trimmed means. Under stayer parallel trends,

$$\left[\mu_{T,1} - \mu_{T,0}^{hi}\right] - \left[\mu_{C,1} - \mu_{C,0}\right] \leq \tau^{SATE} \leq \left[\mu_{T,1} - \mu_{T,0}^{lo}\right] - \left[\mu_{C,1} - \mu_{C,0}\right]. \tag{A.8}$$

The logic is the same as in Lee bounds more generally. Trimming from the lower tail produces the largest admissible pre-treatment mean for stayers and therefore the smallest admissible treatment effect. Trimming from the upper tail produces the smallest admissible pre-treatment mean for stayers and therefore the largest admissible treatment effect.

Once bounds for τ^{SATE} are available, bounds for τ^{ATT} follow immediately from

$$\tau^{ATT} = \tau^{SATE} + (1 - w_0)\,\delta$$

after restricting δ to a plausible range. The two-sided case with treatment-induced exits and entries requires additional structure on both margins and is left for future work in this shorter note.